Thanakrit Jomhong

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Education

Mahidol University Aug 2024 – May 2028
Bachelor of Engineering Program in Computer Engineering *GPA: 3.90*

Experience

Western Digital, Internship Jun 2026 – Aug 2026
Manufacturing Engineer(Machine Learning / Computer Vision), AVI Vision Enhancement

- Built a YOLO instance segmentation pipeline for an Automated Visual Inspection (AVI) system on a live Genesis production line, targeting false-reject reduction in the ML re-inspection workflow.
- Constructed a labeled defect dataset by hand-annotating 1,500+ production images across 6 defect classes using open-source annotation tooling. established labeling documentation standards to ensure annotation consistency and enable dataset expansion.
- Integrated the model into the existing inspection pipeline, reducing manual review workload by 64%.

Head of Sponsorship Coordinator, ClickCamp16 May 2025 – Dec 2025
Mahidol Computer Engineering

- Led sponsorship strategy and relationship management for a university computing camp, secured 130,000 THB in funding.
- Contacted 200+ companies and built structured outreach-tracking workflows to manage the full sponsorship pipeline.

Projects

Banana Ripeness Classification | ResNet34, PyTorch, scikit-learn

- Built a proprietary dataset from scratch by photographing bananas daily over 3+ months across their full ripeness lifecycle, capturing natural progression from fresh to rotten under real conditions.
- Iterated on class taxonomy through multiple relabeling experiments, testing different ripeness stage boundaries to maximize inter-class separability, improving F1-score to 87.5%.

HR ChatBot | n8n selfhost, Docker, Supabase (PostgreSQL), LLM, RAG

- Developed a cross-platform chatbot to automate HR workflows such as leave requests and benefit inquiries through the LINE Messaging API.
- Built a RAG pipeline over HR policy documents with LLM-based intent extraction, handling multiple intent categories end-to-end.

EnergyPulse | Python, SQL, SQLite, dbt, Apache Airflow, BigQuery

- Developed a data ingestion prototype processing 100,000 synthetic energy records in 0.194 seconds in a local in-memory SQLite benchmark.
- Implemented primary-key-based idempotent replay, reprocessing 10,000 existing records with zero additional rows in the prototype experiment.
- Evaluated SQL query optimization through indexed versus non-indexed execution, measuring a 99.9% reduction in median query latency for a selective analytical query.
- Tested data-quality validation against 1,000 deliberately injected invalid records in a 100,000-record synthetic dataset.

Skills

ML / CV: PyTorch, Ultralytics YOLO, OpenCV, scikit-learn, Pandas, NumPy
Systems & Tools: Docker, Git/GitHub, FastAPI, Linux, Supabase, Vercel, n8n
Data Engineering & Analytics: ETL/ELT Pipelines, Data Cleaning & Validation, SQL Query Optimization, Relational Database Design, Data Modeling, API Data Ingestion
English: TOEIC 865

Awards

2nd Place Winner, Super AI Engineer SS5 (AI Innovator Track, Western Region) Summer 2025

Gold Medal, Onsite Presentation, Data Science Project Contest 2023, CMU Sep 2023